

**Task 1:** To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

**Code:**

```
#include <stdio.h>
#include <string.h>
#define MAX 256
int main() {
    char input[MAX];
    while (1) {
        printf("mysh> ");
        fflush(stdout);
        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\n");
            break;
        }
        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
            continue;

        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        printf("You entered: %s\n", input);
    }
    return 0;
}
```

**Output:**

```
● hrushika@hrushikas-MacBook-Air 0SSP % gcc w3s-t2.c -o w3s-t2
● hrushika@Hrushikas-MacBook-Air 0SSP % ./w3s-t1
mysh> exit
Exiting shell...
❖ hrushika@Hrushikas-MacBook-Air 0SSP %
```

**Task 2:** Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close()**

**open()**

syntax: `open(" filename.txt", modes, permissosns)`

Example: `open("First_Prgm.txt", O_CREAT | O_WRONLY, 0644);`

*Table 1: Modes of Open system call*

| Flag     | Purpose                                         |
|----------|-------------------------------------------------|
| O_RDONLY | Open for read only                              |
| O_WRONLY | Open for write only                             |
| O_RDWR   | Open for both read and write                    |
| O_CREAT  | Create file if it does not exist                |
| O_TRUNC  | Truncate (empty) an existing file               |
| O_APPEND | Append data to the end of the file              |
| O_EXCL   | Fail if file already exists (used with O_CREAT) |

read()

Syntax: read(int fd, void \*buf, size\_t count);

Example: read(fd, buffer, 100)

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
int main()
```

```
{
```

```
    char buffer[100];
```

```
    int n;
```

```
    printf("Enter some text: ");
```

```
    n = read(0, buffer, sizeof(buffer) - 1);
```

```
    buffer[n] = '\0'; // Null-terminate the string
```

```
    printf("You entered: %s", buffer);
```

```
    return 0;
```

```
}
```

write()

Syntax: write(int fd, const void \*buf, size\_t count);

Example: write(fd, message, 11);

```
#include <unistd.h>
```

```
int main()
```

```
{
```

```
    write(1, "Hello World\n", 12);
```

```
    return 0;
```

```
}
```

**Code:**

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <fcntl.h>
```

```
#define BUF_SIZE 128
```

```
int main() {
```

```
    int fd1, fd2, n;
```

```
    char buffer[BUF_SIZE];
```

```
    fd1 = open("File1.txt", O_RDONLY);
```

```
    if (fd1 < 0) { perror("open File1.txt"); exit(1); }
```

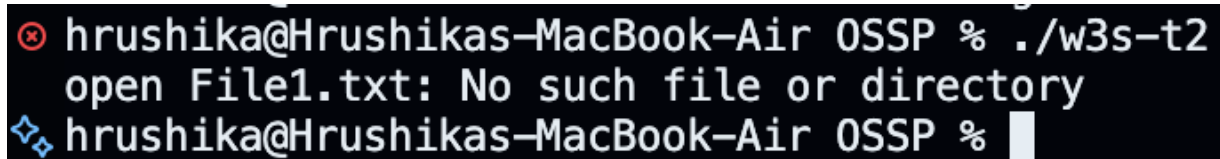
```
    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);
```

```
    if (fd2 < 0) { perror("open File2.txt"); close(fd1); exit(1); }
```

```
    while ((n = read(fd1, buffer, BUF_SIZE)) > 0) {
```

```
        if (write(fd2, buffer, n) != n) {  
            perror("write");  
            close(fd1); close(fd2);  
            exit(1);  
        }  
    }  
    if (n < 0) perror("read");  
  
    close(fd1);  
    close(fd2);  
    printf("File1.txt copied to File2.txt successfully.\n");  
    return 0;  
}
```

Output:



```
❌ hrushika@Hrushikas-MacBook-Air 0SSP % ./w3s-t2  
open File1.txt: No such file or directory  
❖❖ hrushika@Hrushikas-MacBook-Air 0SSP %
```